

- 1 (a) scalar has only magnitude B1
vector has magnitude and direction B1 [2]
- (b) kinetic energy, mass, power all three underlined B1 [1]
- (c) (i) $s = ut + \frac{1}{2} at^2$
 $15 = 0.5 \times 9.81 \times t^2$
 $T = 1.7 \text{ s}$ C1
A1 [2]
- if $g = 10$ is used then -1 but only once on paper
- (ii) vertical component v_v :
 $v_v^2 = u^2 + 2as = 0 + 2 \times 9.81 \times 15$ or $v_v = u + at = 9.81 \times 1.7(5)$
 $v_v = 17.16$ C1
resultant velocity: $v^2 = (17.16)^2 + (20)^2$ C1
 $v = 26 \text{ m s}^{-1}$ A1 [3]
- If $u = 20$ is used instead of $u = 0$ then 0/3
Allow the solution using:
initial (potential energy + kinetic energy) = final kinetic energy
- (iii) distance is the actual path travelled B1
displacement is the straight line distance between start and finish points (in that direction) / minimum distance B1 [2]

- 2 (a) (i) base units of D :